Heuristic Evaluation

Part I: Your Name

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Part II: Project Description

The project I analysed, called "GymTonic", aims to assist gym novices in understanding various exercises, creating a workout plan, and achieve a certified understanding of fitness concepts.

Part III: Evaluation Execution

To conduct the heuristic evaluation of the project under analysis, I scheduled an in-person meeting with the duo responsible for developing the prototype of the mobile version.

The evaluation took place using the low-fi paper prototype with the assistance of both individuals, one portraying the computer and the other acting as the facilitator.

After a brief project introduction by the facilitator, which included an explanation of the identified problem, the proposed solution and the three main tasks, I initiated my evaluation, aiming to autonomously complete all three tasks, focusing on the principles underlying Nielsen's 10 heuristics.

Upon completing the three tasks, I further explored the application, looking for unused functionalities. During this last phase, I posed some questions to the facilitator to clarify to me specific aspects.

During all the evaluation, I decided, with the consent of those present, to record the entire session and to take notes for the compilation of a comprehensive report.
The entire procedure took approximately 45 minutes.

Part IV: List of Violations

Below is the list of identified violations:

1. **H4: Consistency and standards**
    a. **Where:** within the section dedicated to daily training, after selecting "reset", in the confirmation pop-up
    b. **What:** the possible options for the user in the pop-up are unclear
    c. **Why:** In the pop-up, the available options for the user are the following:
        i. "remove all", which is inconsistent with the "reset" button that triggered the pop-up
        ii. "cancel", which may confuse the user as it is unclear whether it prevents the operation of "reset" or actually cancels the daily training
    d. **Severity**: 2

2. **H4: Consistency and standards**
    a. **Where:** within the home, after selecting the badge in the right-top corner of the screen, in the informational pop-up
    b. **What:** the badge related to the “beginner” level makes the user confused
    c. **Why:** the badge implies that a new user is at level 0, but, in the following row, it is specified that the “level 0” badge remains the same from level 0 to level 4, causing confusion for the user
    d. **Severity**: 1

3. **H5: Error prevention**
    a. **Where:** in the section dedicated to daily training
    b. **What:** the removal of an exercise from the daily training made by mistake can’t be stopped
    c. **Why:** once the trash bin button, dedicated to removing the respective exercise from the daily training, is pressed, there is no confirmation pop-up, providing no way for the user to recover a mistakenly deleted exercise
    d. **Severity**: 3

4. **H5: Error prevention**
    a. **Where:** within the section dedicated to daily training, after selecting the swipe-exercise button, in the swipe-exercise pop-up
    b. **What:** the swiping of an exercise in the daily training made by mistake can’t be stopped
    c. **Why:** once the swipe-exercise button is pressed, there is no confirmation pop-up, providing no way for the user to recover a mistakenly swapped exercise
    d. **Severity**: 3

5. **H3: User control and freedom**
    a. **Where:** within the section dedicated to quizzes, after completing a quiz
    b. **What:** review of a quiz once it is successfully completed is not possible
    c. **Why:** after the user successfully completes a quiz, he is informed of his final result, which includes the number of correct and incorrect answers.
    However, there is no way for the user to identify which specific answers were incorrect.
    This possibility is crucial in a project dedicated to education and learning
    d. **Severity**: 4

6. **H6: Recognition rather than recall**
    a. **Where:** within the section dedicated to quizzes, after selecting the info-button, in the informational pop-up
    b. **What:** the badge for the highest level is displayed without mentioning its corresponding name
    c. **Why:** a user who has previously viewed the informational pop-up on the home page knows that a specific name is associated with that badge, but he is forced to remember it or return there to find it out
    d. **Severity**: 1

7. **H4: Consistency and standards**
    a. **Where:** within the section dedicated to daily training
    b. **What:** different procedure from the exercise section for the understanding of how to properly execute an exercise
    c. **Why:** in the exercises section, users can simply tap on the name of an individual exercise to access the connected informational page.
    However, in this particular section, this convenience is removed and replaced by a button
    d. **Severity**: 3

8. **H4: Help and documentation**
    a. **Where:** within the section dedicated to exercises, after selecting a muscular group to train
    b. **What:** the sentence 'tap on exercise for further details', which aims to help the user, is unclear and incomplete
    c. **Why:** when the user reads that sentence, he initially takes a look at an exercise and attempts to tap on the "+" button to get additional information. However he receives the message 'added to your daily training', and without understanding its meaning, he may only complete the task on the second attempt.
    It could be improved by providing information about what this button allows the user to do
    d. **Severity**: 2

9. **H2: Match between system and the real world**
    a. **Where:** within the section dedicated to quizzes
    b. **What:** the symbol "#" as a replacement for the word "numbers"
    c. **Why:** in the sentence, to inform the user about the number of exercises done for a specific muscle group, the word "numbers" is replaced with the symbol "#", however, this is a convention commonly used in mathematics/computer science word and not part of everyday language
    d. **Severity**: 2

10. **H10: Help and documentation**
    a. **Where:** within the section dedicated to quizzes, after selecting the info-button, in the informational pop-up
    b. **What:** the information provided on how to successfully complete a level is not clear
    c. **Why:** in the informational pop-up, it states "complete one quiz by answering correctly to the questions and increase your level" but is never specified what percentage of correct answers out of the total number of questions is required to pass the quiz.
    When completing the quiz, the application sometimes considers it passed with 14/15, while in other instances, it considers it not passed with 6/15.
    This confusion may frustrates the user and for this reason I consider such error significant in an application dedicated to education and learning
    d. **Severity**: 4

11. **H6: Recognition rather than recall**
    a. **Where:** within the section dedicated to daily training
    b. **What:** it is not possible to retrieve to which specific muscle each exercise targets
    c. **Why:** for each exercise included in the daily training, there is no information available regarding the muscle it targets. Even when opening the exercise details, such information is absent. Therefore, the user is forced to remember it or watch the video, understand the muscular group, such as "arms", and then searching by trials and errors among subcategories like "biceps, triceps, …"
    d. **Severity**: 3

12. **H6: Recognition rather than recall**
    a. **Where:** within the section dedicated to exercises, after searching something, in the results display
    b. **What:** when the user searches something, he doesn't see which muscle group the exercises belongs to.
       The information seems to have vanished
    c. **Why:** if the user searches for something generic using the search bar, such as "curl", exercises are displayed, grouped by the muscle group they target; however, this information is not shown in groupings.
       The user is forced to remember, if he knows, the relative muscle group or randomly search for it in the section dedicated to showing all exercises in the application, losing the results of the initial search
    d. **Severity**: 3

13. **H4: Consistency and standards**
    a. **Where:** within the section dedicated to exercises, after selecting one muscular group
    b. **What:** when the user selects a muscle group, like "arms", he receives a list of related muscles, such as "biceps, triceps, …"; however, the way each individual muscle is identified as clickable changes compared to the previous step
    c. **Why:** the main goal for a user in this section of the app is to view a series of exercise related to a selected muscle group.
       However, it could be helpful if, after receiving the list of muscles belonging to the group, he could click on each of them using the same button as before, to open the respective list of exercises
    d. **Severity**: 1

14. **H8: Aesthetic and minimalist design**
    a. **Where:** within the section dedicated to quizzes, after passing a quiz
    b. **What:** the badge that the user receives after passing a single quiz could be avoided
    c. **Why:** using a badge to signify the successful completion of a test could potentially confuse the user, as there is already a level certification system based on badges.
       It could be replaced with a simpler "completed" label or a checkmark
    d. **Severity**: 1

15. **H6: Recognition rather than recall**
    a. **Where:** within the section dedicated to quizzes, in the levels bar
    b. **What:** in the bar indicating the user's level, information about the achieved level and its corresponding name is missing
    c. **Why:** in the quiz section, dedicated to achieving certified knowledge levels, information about the acquired level and the next level to reach is missing.
       This information is only available in an informational pop-up executively accessible from the home, forcing the user to remember it or to navigate there to retrieve the information
    d. **Severity**: 2

16. **H7: Flexibility and efficiency of use**
    a. **Where:** within the section dedicated to daily training
    b. **What:** for more experienced users, seeking to speed up their app usage, it could be beneficial to have a shortcut to save a daily training for later reuse.
    c. **Why:** at the gym, individuals often tend to repeat the same exercise routine for a period of time. A user might want to expedite the loading of the daily training by saving one and reloading it as needed
    d. **Severity**: 1

17. **H9: Help users recognize, diagnose, and recover from errors**
    a. **Where:** within the section dedicated to exercises, after searching something, in the list of results
    b. **What:** If the user searches for an exercise that doesn't exist, either because it's not loaded in the application or due to a typing error, he receive a blank page, preventing them from understanding where the error is
    c. **Why:** If the application doesn't find exercises related to the search, it currently returns an empty list of exercises. Instead, it could display a message to assist the user, possibly suggesting alternative exercises with names similar to the one entered. This would help the user understand and avoid typing errors in the future
    d. **Severity**: 3

18. **H10: Help and documentation**
    a. **Where:** within the section dedicated to quizzes, after selecting the info-button, in the informational pop-up
    b. **What:** the sentence "you can try a quiz any number of times" is not true and may make the user confused
    c. **Why:** in completing a quiz, once it is successfully passed, it is no longer possible to retake the quiz. I recommend reviewing either the functionality or the wording of the statement
    d. **Severity**: 1

19. **H4: Consistency and standards**
    a. **Where:** within the section dedicated to exercises, after searching something, in the results display
    b. **What:** in the list of displayed exercises there is no more the button "+", aside each exercise, to directly add it to user's daily training
    c. **Why:** this option is present when exercises are displayed without filters applied
    d. **Severity**: 3

Part V: Summary and Recommendations

Below is a table summarizing the number of violations, divided by the reference heuristics, found

Heuristic	**# violations**
H1: Visibility of system status	0
H2: Match between system and the real world	1
H3: User control and freedom	1
H4: Consistency and standards	6
H5: Error prevention	2
H6: Recognition rather than recall	4
H7: Flexibility and efficiency of use	1
H8: Aesthetic and minimalist design	1
H9: Help users recognize, diagnose, and recover from errors	1
H10: Help and documentation	2
HN: Non-heuristic issue	0

In my opinion, the application fulfils its main purpose, which is to assist gym users in understanding different types of exercises and their execution. I particularly appreciate how it can be used both by fitness novices to start correctly and by more experienced users who can discover new exercises through the swap function and test their knowledge.

Suggestions I can provide, in addition to working on the heuristics, are to enhance the daily training section, as it could offer more features, making that section more comprehensive.
For example, the application could provide the option to manage the number of sets, repetitions, and associated timing, including relevant pauses, for each exercise.

Secondly, considering that the theme is education and learning, I recommend allowing users to retake a quiz that has already been successfully completed. This way, at their discretion, they can refresh their memory or verify if they have retained the acquired knowledge over time.
Lastly, in the profile section, which is correctly not developed in this low-fi prototype, I recommend allowing users to choose the body type they identify with. This way, they can have a suitable figure right from the home section, providing a more personalized experience.